

[磁滯曲線實驗]

組別

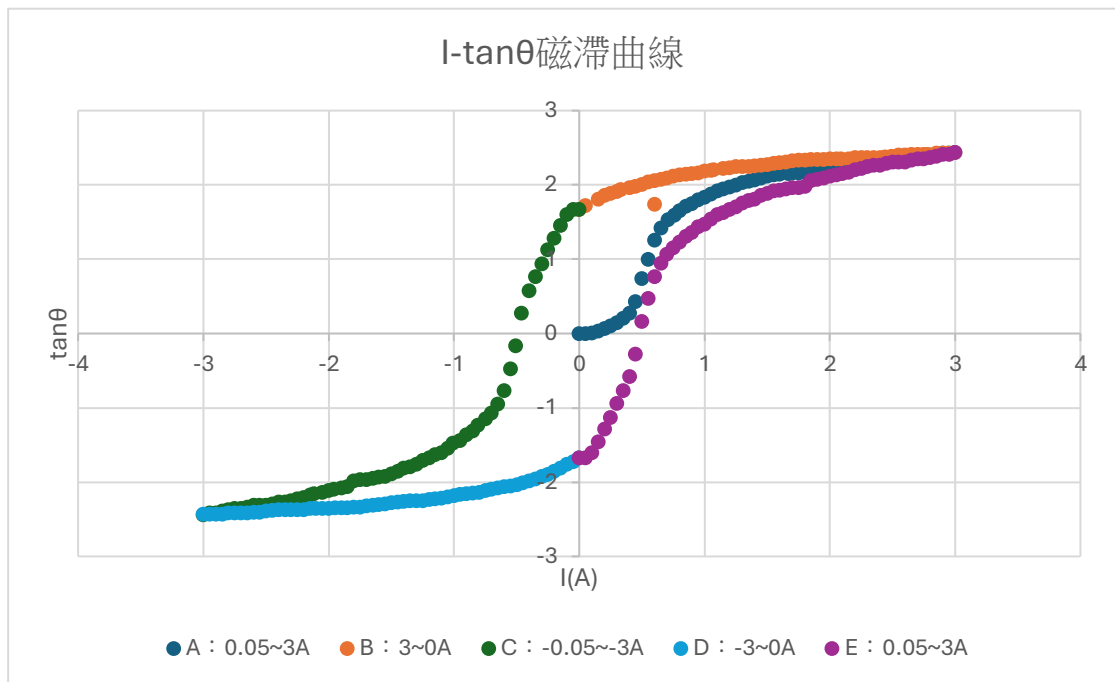
王翊權 E24146107

一、實驗數據

A : 0.05~3A			B : 3~0A			C : -0.05~-3A		
I(A)	θ 平均值	$\tan \theta$	I(A)	θ 平均值	$\tan \theta$	I(A)	θ 平均值	$\tan \theta$
0	0	0	3	67.7	2.438251943	0	59.1	1.670878245
0.05	0	0	2.94	67.6	2.426181858	-0.05	59.1	1.670878245
0.1	0.6	0.010472358	2.9	67.6	2.426181858	-0.1	58	1.600334529
0.15	1.9	0.033173417	2.85	67.6	2.426181858	-0.15	55.5	1.455009029
0.2	3.8	0.066419927	2.78	67.5	2.414213562	-0.2	52	1.279941632
0.25	5.7	0.099813269	2.75	67.5	2.414213562	-0.25	48.4	1.126327107
0.3	8.5	0.149451001	2.7	67.5	2.414213562	-0.3	43.2	0.939062506
0.35	11.5	0.203452299	2.65	67.5	2.414213562	-0.35	37.6	0.770103672
0.4	15.6	0.279205017	2.6	67.4	2.402345703	-0.4	29.9	0.575025504
0.45	23.4	0.432738642	2.55	67.4	2.402345703	-0.46	15.3	0.273569043
0.5	36.4	0.737263585	2.49	67.3	2.39057695	-0.51	-9.1	-0.160174049
0.55	45	1	2.44	67.2	2.378905995	-0.55	-25.2	-0.470564281
0.6	51.6	1.261685998	2.4	67.1	2.367331554	-0.6	-37.5	-0.767326988
0.65	54.8	1.417590366	2.35	67.1	2.367331554	-0.65	-43.5	-0.948964567
0.71	56.9	1.533996902	2.3	67.1	2.367331554	-0.7	-46.8	-1.06489184
0.76	57.9	1.594136594	2.25	67.1	2.367331554	-0.75	-49	-1.150368407
0.8	58.8	1.651196259	2.2	67.1	2.367331554	-0.81	-51	-1.234897157
0.85	59.7	1.711294902	2.15	67	2.355852366	-0.85	-52.7	-1.312687636
0.9	60.2	1.746098431	2.09	67	2.355852366	-0.9	-53.7	-1.361335036
0.95	61	1.804047755	2.05	67	2.355852366	-0.95	-55.2	-1.438811438
1	61.4	1.834129745	2	67	2.355852366	-1.01	-55.9	-1.476993828
1.05	62	1.880726465	1.95	66.9	2.34446719	-1.05	-57	-1.539864964
1.1	62.5	1.920982127	1.9	66.9	2.34446719	-1.1	-58	-1.600334529
1.15	62.8	1.945789558	1.85	66.9	2.34446719	-1.15	-58.5	-1.631851687
1.2	63.1	1.971107679	1.8	66.8	2.333174808	-1.2	-59.1	-1.670878245
1.25	63.4	1.996953856	1.75	66.8	2.333174808	-1.25	-59.6	-1.704458734
1.3	63.8	2.032268347	1.7	66.7	2.321974022	-1.3	-60.4	-1.760318346
1.35	64	2.050303842	1.65	66.6	2.310863654	-1.35	-60.8	-1.789289319
1.4	64.2	2.068599355	1.6	66.5	2.299842547	-1.4	-61.1	-1.811496875
1.45	64.5	2.096543599	1.55	66.4	2.288909564	-1.45	-61.7	-1.857201543
1.5	64.7	2.115516356	1.5	66.3	2.278063586	-1.5	-62.1	-1.888671342
1.55	64.9	2.134771408	1.45	66.2	2.267303512	-1.55	-62.5	-1.920982127
1.6	65	2.144506921	1.4	66.1	2.256628263	-1.6	-62.6	-1.929195568
1.67	65.1	2.154315584	1.35	66	2.246036774	-1.65	-62.8	-1.945789558
1.7	65.2	2.164198288	1.3	66	2.246036774	-1.7	-63	-1.962610506
1.75	65.2	2.164198288	1.25	66	2.246036774	-1.75	-63	-1.962610506
1.8	65.3	2.174155933	1.2	65.9	2.235528	-1.8	-63.2	-1.979663518
1.85	65.8	2.225100911	1.15	65.8	2.225100911	-1.85	-64.1	-2.059418718
1.9	65.9	2.235528	1.07	65.6	2.204487764	-1.9	-64.3	-2.077846524
1.95	66	2.246036774	1.05	65.5	2.194299731	-1.96	-64.5	-2.096543599
2	66.1	2.256628263	1	65.4	2.184189436	-2	-64.7	-2.115516356
2.05	66.2	2.267303512	0.95	65.2	2.164198288	-2.05	-64.9	-2.134771408
2.1	66.3	2.278063586	0.9	65.1	2.154315584	-2.12	-65.1	-2.154315584
2.15	66.4	2.288909564	0.85	65	2.144506921	-2.15	-65.3	-2.174155933
2.2	66.6	2.310863654	0.8	64.9	2.134771408	-2.2	-65.6	-2.204487764
2.25	66.7	2.321974022	0.75	64.7	2.115516356	-2.25	-65.8	-2.225100911
2.3	66.8	2.333174808	0.7	64.5	2.096543599	-2.3	-66	-2.246036774
2.35	66.9	2.34446719	0.65	64.3	2.077846524	-2.35	-66.2	-2.267303512
2.4	66.9	2.34446719	0.6	64.1	2.059418718	-2.4	-66.2	-2.267303512
2.45	67	2.355852366	0.55	63.9	2.041253967	-2.45	-66.4	-2.288909564
2.5	67.1	2.367331554	0.5	63.5	2.005689708	-2.5	-66.6	-2.310863654
2.55	67.1	2.367331554	0.45	63.2	1.979663518	-2.55	-66.6	-2.310863654
2.6	67.1	2.367331554	0.4	63	1.962610506	-2.6	-66.6	-2.310863654
2.65	67.2	2.378905995	0.33	62.7	1.937464506	-2.65	-66.8	-2.333174808
2.7	67.3	2.39057695	0.3	62.4	1.912823577	-2.7	-67	-2.355852366
2.75	67.3	2.39057695	0.25	62.1	1.888671342	-2.75	-67	-2.355852366
2.8	67.4	2.402345703	0.2	61.7	1.857201543	-2.8	-67.1	-2.367331554
2.85	67.5	2.414213562	0.15	61.1	1.811496875	-2.85	-67.3	-2.39057695
2.9	67.6	2.426181858	0.6	60.1	1.7390533	-2.9	-67.5	-2.414213562
2.95	67.6	2.426181858	0.05	59.9	1.725090524	-2.95	-67.5	-2.414213562
3	67.7	2.438251943	0	59.1	1.670878245	-3	-67.7	-2.438251943

D : - 3-0A			E : 0.05~3A		
I(A)	θ 平均 值	$\tan \theta$	I(A)	θ 平均 值	$\tan \theta$
-3	-67.6	-2.426181858	0	-59.1	-1.670878245
-2.95	-67.6	-2.426181858	0.05	-59.1	-1.670878245
-2.9	-67.6	-2.426181858	0.1	-58	-1.600334529
-2.85	-67.6	-2.426181858	0.15	-55.5	-1.455009029
-2.8	-67.5	-2.414213562	0.2	-52	-1.279941632
-2.75	-67.5	-2.414213562	0.25	-48.4	-1.126327107
-2.7	-67.5	-2.414213562	0.3	-43.2	-0.939062506
-2.65	-67.5	-2.414213562	0.35	-37.6	-0.770103672
-2.6	-67.4	-2.402345703	0.4	-29.9	-0.575025504
-2.55	-67.4	-2.402345703	0.45	-15.3	-0.273569043
-2.5	-67.3	-2.39057695	0.5	9.1	0.160174049
-2.45	-67.2	-2.378905995	0.55	25.2	0.470564281
-2.4	-67.1	-2.367331554	0.6	37.5	0.767326988
-2.35	-67.1	-2.367331554	0.65	43.5	0.948964567
-2.3	-67.1	-2.367331554	0.7	46.8	1.06489184
-2.25	-67.1	-2.367331554	0.75	49	1.150368407
-2.2	-67.1	-2.367331554	0.8	51	1.234897157
-2.15	-67	-2.355852366	0.85	52.7	1.312687636
-2.1	-67	-2.355852366	0.9	53.7	1.361335036
-2.05	-67	-2.355852366	0.95	55.2	1.438811438
-2	-67	-2.355852366	1	55.9	1.476993828
-1.95	-66.9	-2.34446719	1.05	57	1.539864964
-1.9	-66.9	-2.34446719	1.1	58	1.600334529
-1.85	-66.9	-2.34446719	1.15	58.5	1.631851687
-1.8	-66.8	-2.333174808	1.2	59.1	1.670878245
-1.75	-66.8	-2.333174808	1.25	59.6	1.704458734
-1.7	-66.7	-2.321974022	1.3	60.4	1.760318346
-1.65	-66.6	-2.310863654	1.35	60.8	1.789289319
-1.6	-66.5	-2.299842547	1.4	61.1	1.811496875
-1.55	-66.4	-2.288909564	1.45	61.7	1.857201543
-1.5	-66.3	-2.278063586	1.5	62.1	1.888671342
-1.45	-66.2	-2.267303512	1.55	62.5	1.920982127
-1.4	-66.1	-2.256628263	1.6	62.6	1.929195568
-1.35	-66	-2.246036774	1.65	62.8	1.945789558
-1.3	-66	-2.246036774	1.7	63	1.962610506
-1.25	-66	-2.246036774	1.75	63	1.962610506
-1.2	-65.9	-2.235528	1.8	63.2	1.979663518
-1.15	-65.8	-2.225100911	1.85	64.1	2.059418718
-1.1	-65.7	-2.214754498	1.9	64.3	2.077846524
-1.05	-65.5	-2.194299731	1.95	64.5	2.096543599
-1	-65.4	-2.184189436	2	64.7	2.115516356
-0.95	-65.2	-2.164198288	2.05	64.9	2.134771408
-0.9	-65.1	-2.154315584	2.1	65.1	2.154315584
-0.85	-65	-2.144506921	2.15	65.3	2.174155933
-0.8	-64.9	-2.134771408	2.2	65.6	2.204487764
-0.75	-64.7	-2.115516356	2.25	65.8	2.225100911
-0.7	-64.5	-2.096543599	2.3	66	2.246036774
-0.65	-64.3	-2.077846524	2.35	66.2	2.267303512
-0.6	-64.1	-2.059418718	2.4	66.2	2.267303512
-0.55	-60.1	-1.7390533	2.45	66.4	2.288909564
-0.5	-60.1	-1.7390533	2.5	66.6	2.310863654
-0.45	-60	-1.732050808	2.55	66.6	2.310863654
-0.4	-60	-1.732050808	2.6	66.6	2.310863654
-0.35	-60	-1.732050808	2.65	66.8	2.333174808
-0.3	-60	-1.732050808	2.7	67	2.355852366
-0.25	-60	-1.732050808	2.75	67	2.355852366
-0.2	-60	-1.732050808	2.8	67.1	2.367331554
-0.15	-59.9	-1.725090524	2.85	67.3	2.39057695
-0.1	-59.9	-1.725090524	2.9	67.5	2.414213562
-0.05	-59.9	-1.725090524	2.95	67.5	2.414213562
0	-59.1	-1.670878245	3	67.7	2.438251943

二、數據分析



抗磁力 $H_c(N/L) = -0.491$ $H_c(N/L) = 0.481$

平均抗磁力約為 0.486 A

殘磁量 $Br(\alpha \mu 0) = 1.671$ $Br(\alpha \mu 0) = -1.671$

三、誤差來源與解釋

有時候磁針的變化相當小，肉眼的讀值誤差會影響 $\tan \theta$ 。

實驗前若消磁程序不夠徹底，會導致磁滯曲線左右不對稱。

實驗室周遭的電子設備或鋼筋結構可能干擾磁場，使 B_E 非定值，造成誤差。

儀器旋鈕壞掉，使電壓回調，因為路徑不可逆而造成誤差